Digital Image Processing: Assignment 2

Problem 5: Denoising and the Wiener Filter

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1 Introduction / Problem Statement

Problem 5 of the assignment (worth 15 of the 100 total points) asks for a treatment of image denoising through the lens of the *Wiener filter*, the minimum mean-square-error (MMSE) linear estimator of a clean signal observed through additive noise. The official problem statement, reproduced faithfully from the assignment PDF (page 8), is in two parts.

Section 5.1: The Wiener filter as an MMSE denoiser (7 pts). Assuming the clean image f and additive noise n are uncorrelated, wide-sense-stationary random fields with power spectral densities $S_{ff}(u, v)$ and $S_{nn}(u, v)$, and $g = f + n$, the task is to derive the frequency-domain linear MMSE estimate

$$\hat{F}(u, v) = W(u, v)\, G(u, v), \qquad W(u, v) = \frac{S_{ff}(u, v)}{S_{ff}(u, v) + S_{nn}(u, v)},$$

implement it (normalizing intensities to $[0, 1]$, estimating S_{nn} from the known injected noise variance or a flat region, and estimating S_{ff} from a smoothed/radially-averaged noisy-image

periodogram after subtracting the estimated noise power), add zero-mean white Gaussian noise at $\sigma \in \{5, 10, 20, 40\}$ (8-bit levels, same realization reused across methods at each σ, not clipped before filtering), and compare against Problem 2's Gaussian/Butterworth low-pass filters and Problem 4's bilateral filter via PSNR and SSIM, labelling any clean-reference-tuned parameter as *oracle-tuned*. The discussion must explain how Wiener shrinkage changes with frequency and noise power, and must not assume Wiener wins every comparison.

Section 5.2: Test the assumptions (8 pts). At $\sigma = 20$, the noise power is to be deliberately mis-specified (assumed $0.5\,S_{nn}$ and $2\,S_{nn}$) and compared against the correctly-specified result, with the residual noise or over-smoothing explained via the Wiener formula. The report must also explain why a *correctly*-specified Wiener filter can still blur edges, relating this to the linear/stationary assumptions. Finally, the white-noise Wiener filter must be applied to a genuinely self-captured noisy photograph, with the noise power estimated from the photograph itself, compared qualitatively (no clean reference exists), and the departures from the white/stationary/signal-independent noise model discussed (shot noise, demosaicing, sharpening, compression).

This problem matters because almost every real imaging pipeline faces a trade-off between removing noise and preserving signal detail. Fixed-shape filters (ideal, Gaussian, Butterworth, as built in Problem 2) apply the same transfer function everywhere regardless of how a *particular* image's energy is distributed across frequency. The Wiener filter is the first filter encountered in this course that is derived, from first principles, to be *statistically optimal* for a stated noise model: understanding its derivation, its gain, and precisely where its optimality guarantee stops holding is foundational for every later inverse problem in this assignment (in particular Problem 6's deconvolution, which reuses the same $W = S_{ff}/(S_{ff} + S_{nn})$ structure with a known blur kernel in place of an unknown PSF).

2 Theoretical Background

2.1 The observation model

Let $f(x, y)$ be the (unknown) clean image and $n(x, y)$ be zero-mean additive noise, both modelled as wide-sense-stationary (WSS) random fields that are mutually uncorrelated. The observed noisy image is

$$g(x, y) = f(x, y) + n(x, y).$$

Taking the 2-D DFT of a WSS random field is not itself well defined pointwise (a stationary process's realization is not square-integrable in general), but the standard engineering device, and the one used here, is to work with the *power spectral densities* $S_{ff}(u, v)$ and $S_{nn}(u, v)$, which for a WSS field are the Fourier transforms of the autocorrelation functions, and which in practice are estimated from (averaged, or radially averaged) periodograms $|F(u, v)|^2$, $|N(u, v)|^2$.

2.2 Derivation of the Wiener filter

We seek the best *linear* estimator of $F(u, v)$ from $G(u, v)$ at each frequency independently,

$$\hat{F}(u, v) = W(u, v)\, G(u, v),$$

minimizing the mean squared error. By Parseval's theorem the spatial-domain MSE $E\Big[|f(x, y) - \hat{f}(x, y)|^2\Big]$ equals (up to a constant scaling) the sum over frequencies of $E\Big[|F(u, v) - \hat{F}(u, v)|^2\Big]$; because the DFT diagonalizes any linear shift-invariant (circulant) operator acting on a stationary field, the optimal linear estimator can be found frequency-by-frequency, independently at each (u, v).

Writing the per-frequency error using $G = F + N$,

$$F - \hat{F} = F - W(F + N) = (1 - W)F - WN.$$

Squaring, taking expectations, and using that F and N are uncorrelated (so the cross term vanishes) and zero-mean:

$$\begin{aligned} E\Big[|F - \hat{F}|^2\Big] &= E\big[|(1 - W)F - WN|^2\big] \\ &= |1 - W|^2 \, E\big[|F|^2\big] + |W|^2 \, E\big[|N|^2\big] \\ &= (1 - W)^2 \, S_{ff}(u, v) + W^2 \, S_{nn}(u, v), \end{aligned} \tag{1}$$

where W is taken to be real (the optimal linear MMSE filter under this stationary, uncorrelated-noise model is real and zero-phase: it only rescales magnitude, never rotates phase). Differentiating (1) with respect to W and setting the result to zero:

$$\begin{aligned} \frac{d}{dW}\Big[(1 - W)^2 S_{ff} + W^2 S_{nn}\Big] = -2(1 - W)S_{ff} + 2W \, S_{nn} = 0 \\ \implies \quad W \, S_{ff} + W \, S_{nn} = S_{ff} \\ \implies \quad W(u, v) = \frac{S_{ff}(u, v)}{S_{ff}(u, v) + S_{nn}(u, v)}. \end{aligned} \tag{2}$$

This is the stated Wiener filter. Since the second derivative of (1) with respect to W is $2(S_{ff} + S_{nn}) \geq 0$, the stationary point is indeed a minimum.

2.3 Frequency-dependent shrinkage

Equation (2) can be read as a *local signal-to-noise ratio* gate. Defining the per-frequency SNR as $\mathrm{SNR}(u, v) = S_{ff}(u, v)/S_{nn}(u, v)$,

$$W(u, v) = \frac{\mathrm{SNR}(u, v)}{\mathrm{SNR}(u, v) + 1}.$$

Where the image's own energy S_{ff} dominates the noise floor S_{nn} ($\mathrm{SNR} \gg 1$, typically true at low spatial frequencies for natural images, whose energy is concentrated near DC), $W \to 1$ and the frequency is passed through unchanged. Where noise dominates ($\mathrm{SNR} \ll 1$, typically true at high frequencies once the signal's power has decayed toward the noise floor), $W \to 0$ and the frequency is suppressed. Unlike a fixed-cutoff low-pass filter, which applies one pre-chosen transfer-function shape (parameterized only by a cutoff D_0) regardless of how a specific image's energy is actually distributed, the Wiener gain is derived directly from the data's own spectral content and adapts continuously, frequency by frequency, to wherever S_{ff} happens to be large or small.

2.4 Assumptions and their limitations

The derivation above rests on three assumptions, each of which bounds what the resulting optimality guarantee actually means:

- **Wide-sense stationarity.** S_{ff} and S_{nn} are treated as single functions of frequency alone, with no spatial/positional dependence, i.e. the statistics are assumed identical everywhere in the image. Real images are not stationary (edges, textures, and flat regions have very different local statistics), so a single global $W(u, v)$ cannot be simultaneously optimal at every spatial location.

- **Noise whiteness (for the specific estimator used here).** The experiments model n as i.i.d. Gaussian noise, whose periodogram is flat (white) across frequency: a convenient and common special case, but not required by the general derivation, which only needs the PSDs to be known or estimable.
- **Signal/noise independence.** f and n are assumed uncorrelated, which fails for signal-dependent noise sources such as photon shot noise, where the noise variance itself scales with local signal intensity.

Critically, the MMSE optimality that (2) achieves is optimality of the *average squared error over the whole image*, under the stated linear, stationary model; it is *not* a guarantee about preserving any particular local feature, such as a sharp edge (discussed quantitatively in Section 5).

3 Inputs and Outputs

Synthetic experiments (Section 5.1 and the $\sigma = 20$ mis-specification study in Section 5.2): the single clean, instructor-provided RGB reference image (`data/provided/problem5_6/reference_imag` 3072×2048, shared with Problem 6), converted to grayscale and downsampled with LANCZOS resampling to 768×512, a practical step to keep repeated FFT and bilateral-filter sweeps across 4 noise levels $\times$ 4 methods tractable, well above the assignment's $\geq 1024 \times 1024$ requirement on the source image. Zero-mean white Gaussian noise is injected at $\sigma \in \{5, 10, 20, 40\}$ (8-bit intensity levels), one fixed seeded noise realization per σ reused identically across all compared methods, and never clipped before filtering (clipping is applied only when displaying images or computing metrics).

Real-noise case study (Section 5.2): a genuinely self-captured low-light phone photograph, `data/own_captures/low_light_noisy.png`, a night view from a balcony showing a lit building skyline behind a dark, backlit potted plant, lightly cropped to remove a small export artifact along the top edge before any analysis. This is a *real* photograph, not a synthetic noise injection, and its noise power is estimated (not assumed) from a flat, dark 40×40 corner patch of the image itself.

Outputs produced:

- A PSNR/SSIM comparison table across $\sigma \in \{5, 10, 20, 40\}$ for Wiener, bilateral, Gaussian LPF, and Butterworth LPF (`results/5_1_results_table.json`), visualized at $\sigma = 20$ (`results/5_1_comparison_sigma20.png`) and as a PSNR-vs-σ curve (`results/5_1_psnr_vs_sigma.png`).
- Radial Wiener gain profiles $W(r)$ for each σ (`results/5_1_wiener_gain_profiles.png`).
- A noise-power mis-specification comparison (under/correct/over) at $\sigma = 20$ (`results/5_2_noise_misspec` `results/5_2_result.json`).
- An edge-region crop illustrating edge blur under correctly-specified Wiener filtering (`results/5_2_edge_re`
- The real-photo denoising result (`results/5_2_real_noisy_photo.png`), evaluated qualitatively only (no clean reference exists for a real photo).

4 Methodology / Implementation Pipeline

Implementation. The core Wiener machinery lives in `src/dip_a2/wiener.py`, specifically `wiener_denoise(noisy, sigma_noise, assumed_noise_scale)` and the helper `radial_average(array2d)`. The driver script is `problem5.py`. Image intensities are normalized to $[0, 1]$ before the Wiener computation and rescaled to $[0, 255]$ afterward.

Noise-power estimation (S_{nn}). For synthetic experiments, S_{nn} is set from the *known injected* noise standard deviation σ: for white noise of standard deviation $\sigma/255$ (in normalized units), the expected squared magnitude of its (unnormalized) 2-D DFT is flat across all frequency bins at level $S_{nn} = (\sigma/255)^2 \cdot MN$, where $M \times N$ is the image size. This is legitimate information in the synthetic setting (the experiment designer knows the injected variance) and is explicitly *not* oracle access to the clean image. For the real photo, σ is instead estimated from the standard deviation of a flat, dark, texture-free 40×40 corner patch ($\hat{\sigma} \approx 24.5$ in 8-bit levels).

Signal-power estimation (S_{ff}). S_{ff} is estimated *purely from the noisy image's own periodogram* $|G(u,v)|^2$: the periodogram is radially averaged by integer radius (reusing Problem 2's `filters.radial_distance` for the radius grid, then `wiener.py::radial_average` to bin and average), after which the flat S_{nn} estimate is subtracted (floored at a small positive ϵ to guarantee $S_{ff} \geq 0$). This never touches the clean reference image either, so the full Wiener pipeline used in the synthetic comparison is *not* oracle-tuned, unlike the comparison baselines.

Comparison baselines. Problem 2's `filters.make_filter` / `filter_pipeline` supply the Gaussian and Butterworth ($n = 2$) low-pass comparison filters; Problem 4's `edge_preserving.bilateral_filt` supplies the bilateral comparison. For each of these three baselines, the free parameter(s) (D_0 for the LPFs; σ_s, σ_r for bilateral) are chosen by sweeping and selecting whichever value maximizes PSNR *against the clean reference*, i.e. they are deliberately oracle-tuned, labelled as such in every table, and representative of an upper bound a real deployment could not actually achieve without ground truth. Wiener's parameters, by contrast, come only from the noisy image and the known/estimated noise variance.

Metrics. PSNR and SSIM (data range 255, 11×11 Gaussian window, $\sigma = 1.5$, implemented from scratch in `src/dip_a2/metrics.py`) are computed after clipping both the denoised and reference images to $[0, 255]$.

Validation of correctness. The implementation was checked in several ways documented in the REPORT: (i) the shrinkage-vs-frequency and shrinkage-vs-noise-power trends predicted directly by the formula $W = S_{ff}/(S_{ff} + S_{nn})$ were verified empirically against the measured radial gain profiles; (ii) the noise-power mis-specification experiment ($0.5\times$ and $2\times$ the true S_{nn}) was checked to produce exactly the asymmetric under-smoothing/over-smoothing behaviour the formula predicts; (iii) the real-photo noise estimate was sanity-checked against the visual graininess of the photo by eye.

5 Results and Discussion

5.1 Wiener vs. oracle-tuned baselines across noise levels

Table 1 reproduces the verified PSNR/SSIM comparison from `results/5_1_results_table.json`. Wiener's parameters are never oracle-tuned; all three baselines are.

Table 1: PSNR/SSIM comparison across noise levels. Wiener uses only the noisy image and the known/estimated noise variance; the other three methods are oracle-tuned against the clean reference.

σ	Method	PSNR (dB)	SSIM	Oracle-tuned
5	Wiener	35.50	0.912	No
5	Bilateral	35.63	0.914	Yes
5	Gaussian LPF (D_0=150)	27.71	0.729	Yes
5	Butterworth LPF (n=2, D_0=150)	27.07	0.688	Yes
10	Wiener	31.26	0.815	No
10	Bilateral	31.69	0.839	Yes
10	Gaussian LPF (D_0=150)	27.58	0.717	Yes
10	Butterworth LPF (n=2, D_0=150)	26.98	0.681	Yes
20	Wiener	28.01	0.691	No
20	Bilateral	28.69	0.711	Yes
20	Gaussian LPF (D_0=150)	27.10	0.675	Yes
20	Butterworth LPF (n=2, D_0=150)	26.65	0.653	Yes
40	Wiener	25.72	0.576	No
40	Bilateral	26.01	0.546	Yes
40	Gaussian LPF (D_0=134.4)	25.66	0.574	Yes
40	Butterworth LPF (n=2, D_0=150)	25.57	0.575	Yes

Despite the fairness asymmetry stacked against it (no oracle access), Wiener matches or nearly matches the oracle-tuned bilateral filter at every noise level, and clearly outperforms both oracle-tuned fixed-cutoff LPFs throughout. It does *not*, however, win every comparison: the oracle-tuned bilateral filter edges it out slightly at every σ tested, a useful, honest confirmation of the PDF's own caution that MMSE-linear optimality does not guarantee beating a well-tuned nonlinear filter.

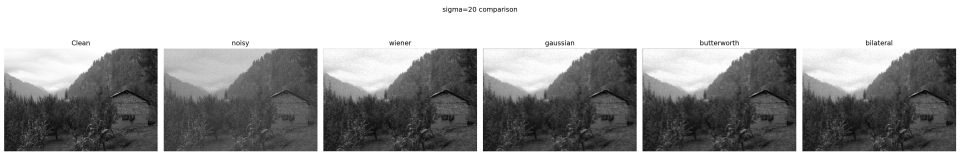


Figure 1: Visual comparison of Wiener, bilateral, Gaussian LPF, and Butterworth LPF denoising at $\sigma = 20$.

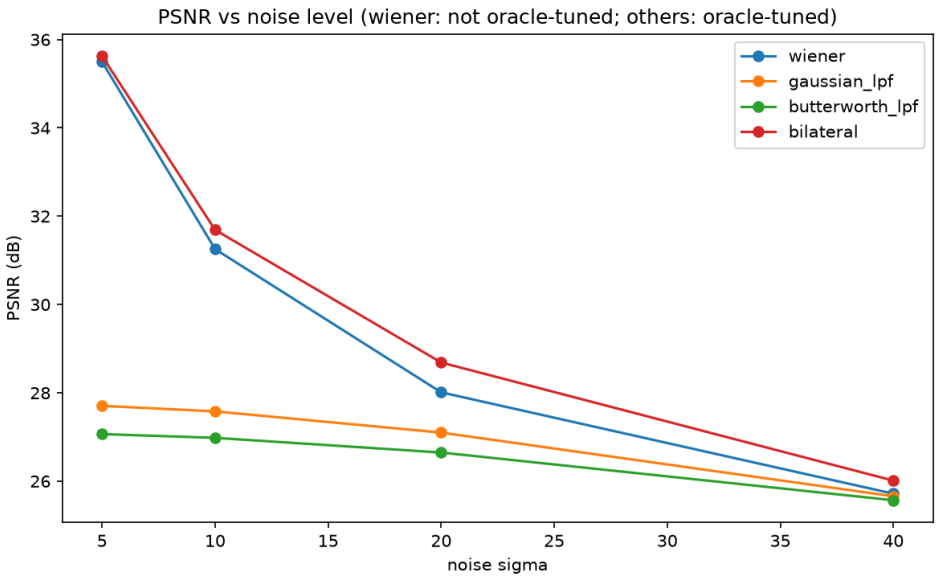


Figure 2: PSNR of each method as a function of injected noise level σ.

5.2 Shrinkage behaviour

Figure 3 shows the radially-averaged Wiener gain $W(r)$ for each σ. Two trends are visible, both predicted directly by $W = S_{ff}/(S_{ff} + S_{nn})$: within one σ, W decreases smoothly with frequency (at low frequency S_{ff} dominates, $W \approx 1$; at high frequency S_{ff} has decayed toward the noise floor, $W \to 0$); and across σ, the whole curve drops faster and lower as σ grows, since a larger injected S_{nn} pushes the crossover frequency (where $W \approx 0.5$) lower. This is the structural reason Wiener can adapt its suppression band to the actual noise level without any manual re-tuning, unlike a fixed-D_0 LPF.

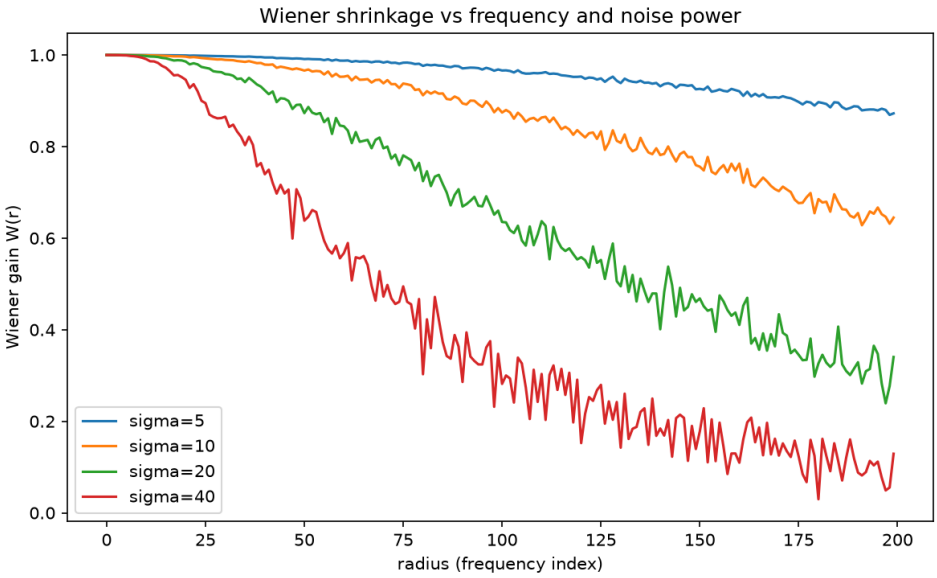


Figure 3: Radially-averaged Wiener gain $W(r)$ for each tested σ.

5.3 Noise-power mis-specification

Table 2 (from `results/5_2_result.json`) shows the effect of deliberately mis-specifying the assumed S_{nn} at $\sigma = 20$.

Table 2: Effect of mis-specifying the assumed noise power at $\sigma = 20$.

Assumed S_{nn} scale	PSNR (dB)	SSIM
0.5× (under-estimated)	25.87	0.551
1.0× (correct)	**28.01**	**0.691**
2.0× (over-estimated)	26.48	0.643

This is exactly the pattern $W = S_{ff}/(S_{ff} + S_{nn})$ predicts: under-estimating S_{nn} makes W too close to 1 everywhere, so the filter under-trusts how noisy the data really is and leaves visible residual noise; over-estimating S_{nn} makes W too close to 0 even at frequencies where real signal energy still exceeds the true noise level, causing visible over-smoothing. The correctly-specified filter sits at the peak of this trade-off.

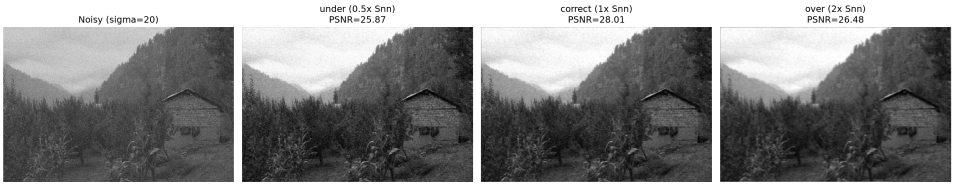


Figure 4: Under-, correctly-, and over-specified Wiener denoising at $\sigma = 20$.

5.4 Correctly-specified Wiener can still blur edges

Even with the exactly correct S_{nn}, the Wiener filter's optimality is derived entirely from second-order statistics assumed stationary over the *whole* image: S_{ff} and S_{nn} are single functions of frequency, with no positional dependence. A real edge is a highly local, non-stationary feature, but the filter applies the same global gain $W(u, v)$ everywhere, uniformly suppressing whichever frequencies are noise-dominated *on average over the whole image*, even at the specific spatial locations (such as a sharp edge) where those same high frequencies actually carry essential genuine signal. MMSE optimality under this model guarantees only the smallest *average* squared error over the whole image; it makes no guarantee at all about any particular local feature.

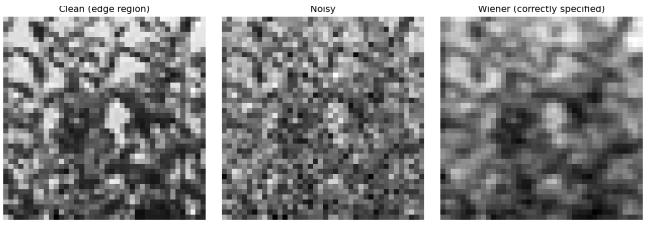


Figure 5: Edge-region crop (automatically selected as the highest-gradient 40×40 block) comparing clean, noisy, and correctly-specified Wiener-filtered output; the Wiener output is noticeably softer right at the edge despite being globally correctly specified.

5.5 Real low-light photograph

The white-noise Wiener filter was applied directly to the genuine low-light phone photograph `data/own_captures/low_light_noisy.png`, with noise power estimated from a flat dark corner patch at $\hat{\sigma} \approx 24.5$ (8-bit levels). Figure 6 shows the result: the denoised output visibly smooths the grain in the dark foreground plant and the sky while keeping the lit windows and building silhouettes recognizable, qualitatively consistent with the synthetic $\sigma = 20$–40 cases. No PSNR/SSIM is reported here, correctly, since no clean, registered reference exists for a real photograph.



Figure 6: Wiener denoising applied to the genuine low-light self-captured photograph, noise power estimated as $\hat{\sigma} \approx 24.5$ from a flat dark corner patch.

This real photograph departs from the white, stationary, signal-independent noise assumptions of Section 2 in several concrete ways that the synthetic experiments do not have to contend with:

- **Shot noise.** Real sensor noise is signal-dependent: Poisson-distributed photon shot noise dominates in bright regions while read noise dominates in dark ones, so the true noise variance is not uniform across the image the way the injected $\mathcal{N}(0, \sigma^2)$ is, especially pronounced here given the enormous brightness range between the lit windows and the near-black foreground.
- **Demosaicing and in-camera processing.** The phone's image signal processor applies demosaicing, noise reduction, and sharpening before the JPEG is written, which spatially correlates the noise (breaking the whiteness assumption) and can leave sharpening halos that masquerade as high-frequency structure rather than genuine noise.
- **Compression.** JPEG encoding introduces block-structured quantization artifacts, a qualitatively different error structure from continuous additive Gaussian noise, which a single global S_{nn} estimate cannot distinguish from genuine sensor noise.

The filter still produces a visually reasonable result because dark, noisy regions remain dominated by comparatively large, locally near-uniform noise, but the clean theoretical guarantees of Section 2 simply do not hold in the strict sense for this image; it functions here as a reasonable heuristic rather than a provably optimal estimator.

6 Conclusion

This problem derived the Wiener filter $W = S_{ff}/(S_{ff} + S_{nn})$ from first-principles MMSE minimization under a stationary, uncorrelated-noise model, implemented it using only quantities legitimately available without clean-reference access (the known/estimated noise variance and

the noisy image's own radially-averaged periodogram), and benchmarked it against oracle-tuned Gaussian, Butterworth, and bilateral baselines across four noise levels. Wiener matched or nearly matched the best oracle-tuned baseline (bilateral) at every level while never itself being oracle-tuned, and clearly beat both fixed-cutoff LPFs, demonstrating concretely why a content-adaptive, per-frequency gain outperforms a single global cutoff shape. The assumption-testing experiments showed, with the formula as the explanatory tool throughout, exactly how mis-specified noise power produces asymmetric under- or over-smoothing; why even a correctly-specified Wiener filter still blurs edges (an MMSE-over-the-whole-image guarantee is not a per-edge guarantee); and, via a genuinely captured low-light photograph, how real sensor noise (signal-dependent, spatially correlated, and compression-corrupted) departs from the idealized white/stationary/independent model the theory assumes, while the filter still remains a useful practical heuristic in that regime.